EXP 17: Implement Mobile Price Prediction using appropriate machine learning algorithm

CODING:

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# Mobile Price Prediction using KNN

# Dataset: [RAM in GB, Storage in GB, Price Category]
data = [
  [2, 32, "Low"],
  [3, 32, "Low"],
  [4, 64, "Medium"],
  [6, 64, "Medium"],
  [8, 128, "High"],
  [12, 256, "High"]
]

# New mobile details
test = [8, 128]

# Calculate Euclidean distance
distances = []

for row in data:
  distance = ((test[0] - row[0]) ** 2 +
        (test[1] - row[1]) ** 2) ** 0.5

  distances.append((distance, row[2]))
```

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# Sort distances
distances.sort()

# Select 3 nearest neighbors
k = 3
neighbors = [item[1] for item in distances[:k]]

# Find majority class
prediction = max(set(neighbors), key=neighbors.count)

print("MOBILE PRICE PREDICTION")
print("----------------------")
print("RAM:", test[0], "GB")
print("Storage:", test[1], "GB")
print("Predicted Price Category:", prediction)
```

OUTPUT:

